

LESSON 2-9 • SESSION 1 • GUIDED NOTES**Finding Mean Absolute Deviation (MAD)**

Learning target – I can find the mean absolute deviation of a data set and explain what it measures. · 6.SP.B.5c

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
mean absolute deviation (MAD) <i>desviación media absoluta</i>	The ____ of the distances from the mean.	distances 4, 0, 1, 3: MAD 2
deviation <i>desviación</i>	How ____ one value is from the mean.	mean 19, value 23: 4
absolute value <i>valor absoluto</i>	A number's distance from 0; it is never ____.	$ -5 = 5$
spread <i>dispersión</i>	How far ____ the values are.	small MAD = tight set

KEY IDEA*complete each sentence*

- Find the mean, then each value's ____ from the mean.
- The MAD is the ____ of those distances.
- A small MAD means the values are ____ to the mean.

WORKED EXAMPLE*fill in the blanks as we solve*

Four plants grew 3, 7, 8 and 10 cm. Find the MAD.

Step 1 Mean: $28 \div 4 =$ ____**Step 2** Distances from 7: 4, ____, 1, 3**Step 3** Add the distances: _____. Divide by 4.**Answer** MAD = ____ cm**YOU TRY***use the example as a model***1.** The mean is given. Find the MAD.**a.** 2, 4, 6, 8 (mean 5) _____**b.** 8, 14, 16, 22 (mean 15) _____**2.** The mean of 4, 6, 8, 10, 12 is 8. Find the MAD.*Show your work.*

MAD _____

SUM IT UP*in your own words*

The MAD tells me ____ because ____.

LESSON 2-9 • SESSION 2 • GUIDED NOTES

Comparing Data Sets Using MAD

Learning target – I can compare two data sets that share a mean by comparing their mean absolute deviations. • 6.SP.B.5c

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
variation <i>variación</i>	How _____ the values are from each other.	larger MAD = more variation
consistent <i>consistente</i>	Values stay _____ together; the spread is small.	small MAD = consistent
MAD <i>desviación media absoluta</i>	The mean _____ of the values from the mean.	MAD 1.5 versus MAD 4

KEY IDEA*complete each sentence*

- Two sets can have the same mean and _____ MADs.
- The set with the _____ MAD is steadier (more consistent).

WORKED EXAMPLE*fill in the blanks as we solve*

Baker A made 8, 9, 11, 12 cookies per tray. Baker B made 4, 8, 12, 16. Both means are 10. Who is steadier?

Step 1 A's distances: 2, 1, 1, 2. $MAD = 6 \div 4 = \underline{\hspace{1cm}}$ **Step 2** B's distances: 6, 2, 2, 6. $MAD = 16 \div 4 = \underline{\hspace{1cm}}$ **Answer** Baker is steadier: $1.5 < 4$.**YOU TRY***use the example as a model*

- Both sets have a mean of 30. Find each MAD.
 - Set 1: 26, 28, 32, 34 _____
 - Set 2: 20, 24, 36, 40 _____
- Machine A fills bottles with mean 500 mL and MAD 2 mL. Machine B: mean 500 mL, MAD 14 mL. Which is more consistent?

*Show your work.***Machine** _____**SUM IT UP***in your own words*

When two sets have the same mean, I compare _____ because _____.
